1. Clinical Question & Core Hypothesis

Clinical Question

In breast cancer, AR signaling can substitute for or antagonize ER-driven programs, but it is unclear under what chromatin contexts AR activation actually drives endocrine therapy resistance. 2) Tier 1 path(s) used: the single provided path linking androgen→AR activation and the observation that KDM6A is mutated in a relevant cancer model (BM18) — I used AR and KDM6A as anchors because KDM6A loss alters H3K27me3 chromatin repression and could enable ectopic AR transcriptional programs. 3) Tier 2 support: no formal Tier 2 paths were provided; however, the TCGA-BRCA context (available hormone-receptor signaling measures) motivates testing endocrine therapy resistance as the clinical readout. 4) Endpoint choice: therapy_response (endocrine therapy resistance) was chosen because the mechanistic question asks when AR activity functionally bypasses ER dependence, which is most directly assessed by treatment benefit/failure. 5) Deliberate exclusions: I excluded ESR1/PGR and other immune/metabolic candidates because they were not in the supplied path and including them would violate the 2-gene anchor limit and dilute the mechanistic focus. I selected the imaging feature (TSR) because stromal abundance modulates chromatin–niche interactions and may permit AR-driven reprogramming in KDM6A-altered tumors?

Working Hypothesis

In ER-positive breast cancer, loss-of-function of KDM6A reconfigures chromatin by reducing H3K27me3 at AR-associated regulatory regions, enabling androgen receptor (AR) to drive an alternative transcriptional program that supports proliferation and survival when estrogen signaling is inhibited by endocrine therapy. This AR-driven reprogramming is more pronounced in tumors with high stromal-to-tumor ratio (TSR), where stromal signals cooperate with chromatin remodeling to facilitate AR access to non-canonical targets. Clinically, ER+/AR-high, KDM6A-mutant, TSR-high tumors will exhibit reduced benefit from endocrine therapy (shorter therapy_response duration and worse progression-free interval [PFI]/overall survival [OS]) compared with AR-high, KDM6A-wild-type, TSR-low tumors. The working model predicts: (1) AR target gene signatures shift toward AR-driven programs in KDM6A-deficient contexts; (2) chromatin assays will show AR binding at hitherto repressed loci with concurrent local chromatin opening; (3) functional AR activity under endocrine therapy will be preserved or enhanced in KDM6A-deficient cells, particularly in a TSR-mimicking microenvironment. This will be tested through integrated computational pathology (quantifying TSR, AR/KDM6A status) with clinical therapy outcome data, complemented by in vitro/in vivo experiments using ER+ cell lines with KDM6A knockout and AR activation, plus chromatin and transcriptional profiling to establish causality.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	AR, KDM6A
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

AR: AR activation and the observation that KDM6A is mutated in a relevant cancer model (BM18) — I used AR and KDM6A as anchors because KDM6A loss alters H3K27me3 chromatin repression and could enable ectopic AR transcriptional programs.

KDM6A: KDM6A is mutated in a relevant cancer model (BM18) — I used AR and KDM6A as anchors because KDM6A loss alters H3K27me3 chromatin repression and could enable ectopic AR transcriptional programs.

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions